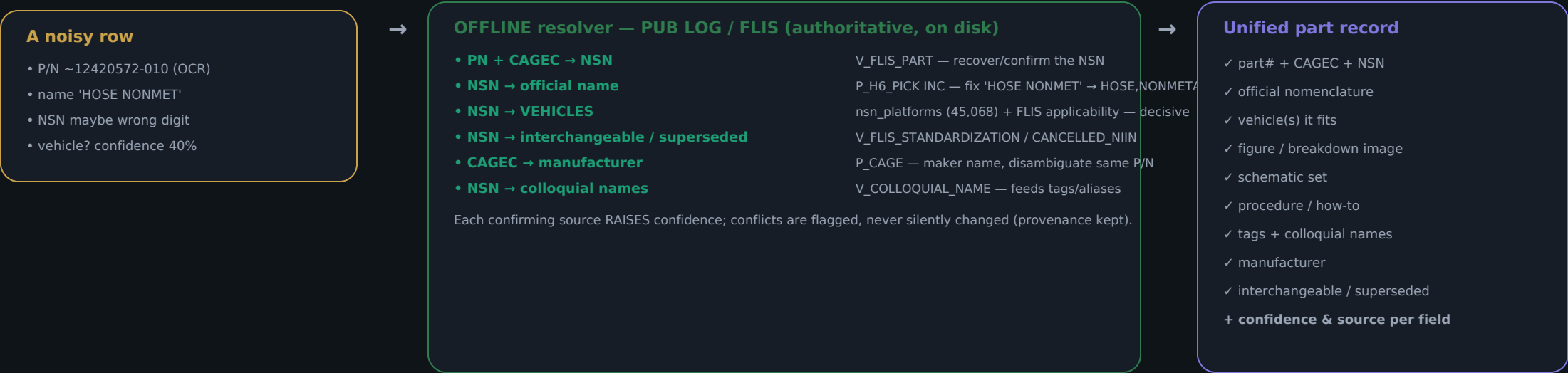


Cross-reference & verification engine (proposal)

Take every OCR-noisy / low-confidence row and decisively resolve it — part ↔ NSN ↔ vehicle ↔ figure/schematic/photo/procedure/tags.



Residual unknowns only → OPTIONAL internet pass (cached, gated, ITAR/copyright-aware)

What FLIS can't resolve goes to an optional online cross-reference for PUBLIC data only — NSN catalogs, manufacturer cross-reference, colloquial names. Results are CACHED into a sidecar so the offline system keeps them; never fetched at runtime, never controlled technical data. Anything still unresolved → the Part# review queue (human-confirmed).

Hard caveats

- Offline-first: FLIS does ~all of it; the internet is optional + cached + gated, NEVER a runtime dependency, and ITAR/copyright-aware (no controlled CAD/tech data).
- Many-to-many: one part fits many vehicles; one NSN has many P/Ns; superseded chains — modeled as links, not a single value.
- OCR residue will never fully zero out → the review queue is the backstop. Every repaired field keeps its source + confidence (auditable, reversible).
- Scale: a host-side batch over the 16 GB PUB LOG + the index (can't run in the sandbox). Sidecar xref.db; the main index is never written (R1/R6).